ADAPTIVE RETRIEVAL ORCHESTRATION FOR RAG USING THE CARO FRAMEWORK IN LARGE LANGUAGE MODELS

Submitted in partial fulfillment for the award of the degree of

Master of Computer Applications

by

SAMPRATI (25MCA1059)



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I hereby declare that the thesis entitled "**ADAPTIVE RETRIEVAL ORCHESTRATION FOR RAG USING THE CARO FRAMEWORK IN LARGE LANGUAGE MODELS**" submitted by me for the award of the degree of **Master of Computer Applications**, Vellore Institute of Technology, Chennai, is a record of bona fide work carried out by me under the supervision of the project guide.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either part or in full, for the award of any other degree or diploma in this institute or any other institute or university. All external concepts, algorithms, datasets, software libraries, and research contributions used in this work have been acknowledged through appropriate references.

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Signature of the Guide:
Name: Dr. Rabindra Kumar Singh
Date:

Signature of the Internal Examiner

Name:

Date:

Signature of the External Examiner

Name:

Date:

Approved by the Head of Department,
Master of Computer Applications
Name: Dr.Graceline Jasmine S
Date:

(Seal of SCOPE)

ABSTRACT

Retrieval-Augmented Generation (RAG) combines information retrieval with Large Language Model generation to improve responses to knowledge-intensive questions. Although RAG can reduce dependence on knowledge stored only in model parameters, many practical systems apply a fixed retrieval process to every query. Such a design does not distinguish between direct questions, semantic retrieval needs, exact lexical searches, multi-hop questions, and conversational follow-up questions that depend on previous context.

This thesis presents CARO, a Context-Aware Adaptive Retrieval Orchestrator designed to make retrieval decisions query-dependent. CARO introduces an orchestration layer between the user query and the retrieval/generation components. The framework first analyzes the query, identifies its question type, independently determines whether external retrieval and conversational memory are required, and then selects an appropriate execution strategy. The architecture is deliberately modular so that established retrieval and generation components can be reused while the orchestration logic remains explainable and testable.

The current version is made up of eight modules. The first module deals with the pre-processing of the data-set and documents. The second module sets the RAG baseline using the BGE-M3 embeddings, FAISS dense retrieval and Gemma 3 4B generation via Ollama. Module 3 picks deterministically one of four possible outcomes, namely DIRECT, RETRIEVE, MEMORY, and RETRIEVE AND MEMORY, from the adaptive query router. Module 4 selects one of five possible strategies, that is, DIRECT, DENSE, SPARSE, HYBRID and MEMORY AND RETRIEVAL depending on the result of routing. The fifth to eighth modules include the retrieval/relevance filtering, temporary memory usage, sufficiency check via iterative retrieval and the respective generation and evaluation.

The HotPotQA dataset will be used as the main dataset to enable multi-hop reasoning over the supporting documents. QReCC may be kept in case there are some future works involving conversational dynamics and memory. Currently, the stage of the project involves HotPotQA data pre-processing pipeline, FAISS index (5,000 documents), nine routing/regression tests for M3, and eight strategy selection tests for M4.

Current thesis states the difference between the success of implementation and planned experiments. Mathematical models of dense similarity, BM25, RRF (Reciprocal Rank Fusion), Routing Confidence, Coverage Evidence, Sufficiency Evidence, Precision and Coverage Retrieval, MRR (mean reciprocal rank), NDCG (normalized discounted cumulative gain), Exact Match, F1, Latency, Context Size and Retrieval Efficiency are introduced. Main research hypothesis suggests that orchestration can improve the quality-efficiency tradeoff of RAG by avoiding redundant retrieval; selecting retrieval methods depending on the features of queries; using contextual memory where it is needed; and conducting further retrievals only when there is not enough evidence.

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Place: Chennai

Date: SAMPRATI

TABLE OF CONTENTS

CONTENTS ..... **vii**

LIST OF FIGURES ..... **xii**

LIST OF TABLES ..... **xiii**

LIST OF ACRONYMS ..... **xvi**

CHAPTER

INTRODUCTION

1.1 Introduction and Context 1

1.2 Motivation 2

1.3 Research Problem 2

1.4 Research Questions 3

1.5 Objectives 3

1.6 Contributions 3

1.7 Scope and Boundaries 4

1.8 Thesis Organization 4

CHAPTER 2

BACKGROUND AND THEORETICAL FOUNDATIONS

2.1 Large Language Models 5

2.2 Information Retrieval 5

2.3 Dense Retrieval 6

2.4 Sparse Retrieval and BM25 6

2.5 Retrieval-Augmented Generation 7

2.6 Vector Similarity and FAISS 7

2.7 Multi-Hop Question Answering 8

2.8 Conversational Retrieval 8

2.9 Evidence Grounding 9

CHAPTER 3

LITERATURE REVIEW

3.1 Evolution of RAG 10

3.2 Fixed RAG Systems 12

3.3 Adaptive Retrieval 13

3.4 Conversational RAG 15

3.5 Multi-Hop RAG 16

3.6 Retrieval Fusion and Reranking 17

3.7 Research Gap 19

3.8 Positioning of CARO 19

CHAPTER 4

PROBLEM DEFINITION AND METHODOLOGY

4.1 Formal Problem Definition 21

4.2 Research Design 21

4.3 Experimental Variables 22

4.4 Datasets 22

4.5 Fair Comparison Protocol 23

4.6 Reproducibility 23

4.7 Threats to Validity 24

CHAPTER 5

CARO FRAMEWORK AND SYSTEM DESIGN

5.1 Design Philosophy 25

5.2 Overall Architecture 27

5.3 Module 1 – Dataset Processing 27

5.4 Module 2 – Fixed Baseline RAG 27

5.5 Module 3 – Adaptive Query Router 28
5.6 Module 4 – Strategy Selection 28
5.7 Module 5 – Retrieval and Reranking 29
5.8 Module 6 – Temporary Memory 30
5.9 Module 7 – Evidence Sufficiency 30
5.10 Module 8 – Generation and Evaluation 31
5.11 Data Flow and Interfaces 31

CHAPTER 6

ALGORITHMS AND MATHEMATICAL FORMULATION

6.1 Notation 32
6.2 Routing Equations 32
6.3 Dense Retrieval 33
6.4 BM25 33
6.5 Reciprocal Rank Fusion 33
6.6 Relevance Filtering 34
6.7 Evidence Coverage 34
6.8 Evidence Sufficiency 34
6.9 Iterative Retrieval 35
6.10 Evaluation Metrics 35
6.11 Efficiency Metrics 36

CHAPTER 7

IMPLEMENTATION AND EXPERIMENTAL SETUP

7.1 Software and Hardware Environment 38
7.2 Dataset Processing 38
7.3 Baseline Implementation 39
7.4 Routing Implementation 39

7.5 Strategy Selection Implementation 39

7.6 Testing Strategy 40

7.7 Experimental Procedure 40

7.8 Reproducibility Checklist 41

CHAPTER 8

RESULTS, ANALYSIS AND DISCUSSION

8.1 Current Implementation Results 42

8.2 Routing Validation 44

8.3 Strategy Validation 45

8.4 Baseline Verification 47

8.5 Proposed Baseline-versus-CARO Experimental Matrix 48

8.6 Error Analysis 49

8.7 Discussion 50

CHAPTER 9

LIMITATIONS, SECURITY, ETHICS AND RELIABILITY

9.1 Current Limitations 51

9.2 Data Limitations 51

9.3 Hallucination and Grounding 52

9.4 Privacy and Temporary Memory 52

9.5 Security Considerations 52

9.6 Reproducibility and Reliability 53

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 Conclusion 55

10.2 Academic Significance 56

10.3 Practical Significance 56

10.4 Future Work 56

REFERENCES – APA FORMAT 58

APPENDICES 60

LIST OF FIGURES

- Figure 1. Conventional Fixed RAG Pipeline 2
- Figure 2. CARO Adaptive Orchestration Pipeline 19
- Figure 3. CARO-RAG Adaptive Retrieval and Evidence Validation Architecture 26
- Figure 4. Evidence Sufficiency and Iterative Retrieval 35
- Figure 5. Controlled Baseline-versus-CARO Evaluation 39
- Figure 6. Module-Level CARO Architecture 49
- Figure 7. Query Routing Decision Flow 43
- Figure 8. Dense and Sparse Retrieval Fusion 48
- Figure 9. Temporary Memory Retrieval 46
- Figure 10. Experimental Error-Analysis Workflow 49

LIST OF TABLES

- Table 1. CARO Module Completion Status 27
- Table 2. Dataset Characteristics and Research Roles 22
- Table 3. Query Types Supported by the Router 29
- Table 4. Routing States 44
- Table 5. Retrieval Strategies 45
- Table 6. Mathematical Notation 32
- Table 7. Software Environment 38
- Table 8. Module-Level Validation 40
- Table 9. Evaluation Metrics 43
- Table 10. Baseline-versus-CARO Experimental Matrix 49
- Table 11. Threats to Validity and Mitigation 24
- Table 12. Limitations and Future Improvements 54

LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

Term	Meaning
CARO	Context-Aware Adaptive Retrieval Orchestrator
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
AQR	Adaptive Query Router
BM25	Best Matching 25
FAISS	Facebook AI Similarity Search
MRR	Mean Reciprocal Rank
NDCG	Normalized Discounted Cumulative Gain
EM	Exact Match
F1	F1 score
RRF	Reciprocal Rank Fusion
q	User query
D	Document collection
d	Individual document
K	Number of retrieved documents
M	Temporary memory
τ	Decision threshold
Rel	Relevance score
Cov	Evidence coverage
Ans	Answerability
Latency	Elapsed inference time
Context size	Number of tokens supplied to the generator